

ELEC5660 Project 1 Phase 2 Report

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April 4, 2026

1 Introduction

This report presents the implementation of the A* (A-Star) path planning algorithm for 3D grid-based navigation. The algorithm plans collision-free paths in a three-dimensional environment with obstacles.

2 A* Implementation Details

2.1 State Representation

Each state in our A* implementation is represented as a 3D grid coordinate:

$$s = (x, y, z) \quad \text{where} \quad x, y, z \in \mathbb{Z} \quad (1)$$

The grid uses 1-based indexing with bounds:

$$1 \leq x \leq \max_x, \quad 1 \leq y \leq \max_y, \quad 1 \leq z \leq \max_z \quad (2)$$

For visualization and execution, grid indices are converted to cell-center coordinates:

$$p_{\text{world}} = [x - 0.5, y - 0.5, z - 0.5] \quad (3)$$

2.2 Neighbor Expansion

We use 6-connected connectivity for neighbor expansion, allowing movement along the three principal axes:

$$\mathcal{N}(s) = \{s + \delta \mid \delta \in \Delta\} \quad (4)$$

where

$$\Delta = \{(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)\} \quad (5)$$

2.3 Heuristic Function

We employ the Manhattan distance as our heuristic function, which is admissible and consistent for 6-connected grids:

$$h(s_1, s_2) = |x_1 - x_2| + |y_1 - y_2| + |z_1 - z_2| \quad (6)$$

2.4 Cost Function

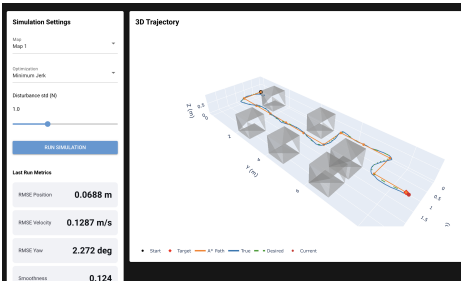
The total cost function follows the standard A* formulation:

$$f(s) = g(s) + h(s) \quad (7)$$

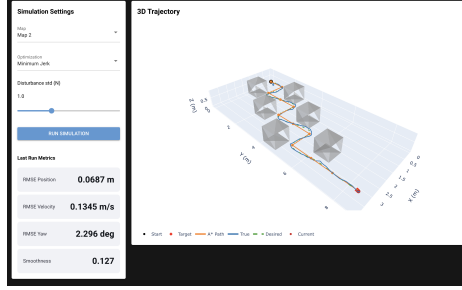
where:

- $g(s)$: Actual cost from start to state s (number of steps)
- $h(s)$: Heuristic estimate from s to goal

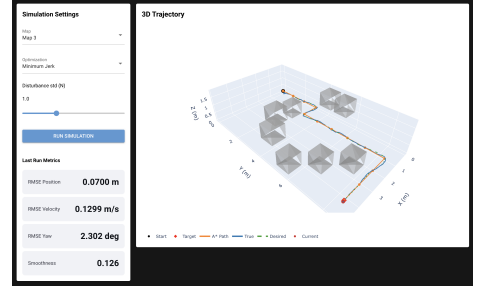
3 Statistics



(a) Planned path and trajectory for Map 1



(b) Planned path and trajectory for Map 2



(c) Planned path and trajectory for Map 3

Figure 1: Planned paths and trajectories for Maps 1–3

Table 1: Path Planning Statistics

Metric	Map 1	Map 2	Map 3
RMSE Position (m)	0.0688	0.0687	0.0700
RMSE Velocity (m/s)	0.1287	0.1345	0.1299
RMSE Yaw (rad)	0.03965	0.04007	0.04018
Smoothness Score	0.124	0.127	0.126